KUNYUAN HU

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Education

University of Toronto Mississauga **Expected May 2029**
Honours Bachelor of Science, Computer Science and Statistics (Double Major) *Mississauga, ON*

Experience

Research Assistant **Oct 2026 – Present**
Dr. Antonio Vivone, University of Toronto *Toronto, ON*

- Audited the dataset and supplementary materials of a multi-classroom education study ahead of manuscript submission, delivering a report of **15 findings** ranked by severity.
- Reproduced the study's **mixed-effects model**, effect sizes and **MANCOVA** from raw data, reconstructing the undocumented classroom groupings needed to verify the reported results.
- Identified two interaction tests mislabelled as omnibus tests, and recomputed the correct **joint Wald tests**.
- Continuing with the project as a **summer research assistant in 2027**.

Server, Part-Time **Dec 2022 – Mar 2024**
The Bradley Gracious Retirement Living *Ottawa, ON*

- Served residents daily in an assisted living community for **15 months** alongside full-time studies, balancing speed of service against individual dietary needs and accessibility requirements.

Projects

Kinvue — Caregiver Check-In with Camera-Based Vitals **Sep 2026 – Present**
Electron, React, TypeScript, Tailwind CSS, Vitest, Presage SmartSpectra SDK

- Building an **Electron + React** desktop app that reads pulse, breathing and HRV from a 30-second webcam capture and flags days that differ from the older adult's own **14-session baseline**.
- Designed a rule-based scoring engine in framework-free TypeScript, covered by **1,100+ Vitest tests**, where every fired rule states its actual values so a caregiver can see why a day was flagged.
- Verified with **Wireshark** that the vitals SDK sends a ~55 kB licence meter rather than video, and encrypted local history with OS key storage and passphrase-protected exports (**AES-256-GCM**).

ClipFarm — Automated Volleyball Highlight Generator **Mar 2026 – Present**
Python, FastAPI, Next.js, TypeScript, YOLOv8-pose, Roboflow, PostgreSQL, Docker, Celery

- Built the rally-detection stage of a highlight pipeline: a custom-trained **Roboflow** ball detector bounds rallies by trajectory, and **audio cheer detection** plus rally-shape scoring rank which rallies become highlights.
- Classified player actions with **YOLOv8-pose** and cut final clips with **FFmpeg**, offloading GPU inference to **Modal** with content-hash caching so repeat uploads of the same footage skip inference entirely.
- Contributed to the **Next.js + TypeScript** frontend — filterable clip grid and lazy-loaded thumbnails — served by **FastAPI** with **Celery + Redis** job distribution over **PostgreSQL**, containerized with **Docker Compose**.

Imposter — Real-Time Multiplayer Party Game **2026**
React, Node.js, WebSockets, Vite

- Built a real-time game that keeps player phones in sync through timed clue and voting phases, using a **Node WebSocket server** as the authoritative source of room state.
- Implemented **four-letter room codes**, turn sequencing, and phase transitions, handling reconnects and mid-game joins without desyncing clients.

Medi-Cal Scheduling — Surgeon–Patient Booking System **2024**
Python, Django, SQLite

- Built a full-stack **Django** booking app around a patient-facing calendar rather than a form queue, with booking and cancellation flows backed by a relational schema in **SQLite**.

Technical Skills

- **Programming Languages:** Python, TypeScript, JavaScript, Java, SQL, HTML/CSS
- **Frameworks:** React, Next.js, Node.js, Electron, Django, FastAPI, Celery, Tailwind CSS
- **ML & Data:** Mixed-effects models, MANCOVA, YOLOv8-pose, Roboflow, OpenCV, PostgreSQL, SQLite
- **Tools:** Git, Docker, FFmpeg, WebSockets, Redis, Modal, Vite, Vitest, Wireshark, REST APIs
- **Languages:** English, French (**DELF B1**), Mandarin